

An Exact Pressure-Multiplicity Refinement for Simple Zeros of the Riemann Zeta Function

AMTOPA and contributors

August 12, 2026

Abstract

We record a short combinatorial refinement of the shifted-block averaging step in the positioned-pressure certificate of `sxuff/zeta-positioned-pressure`. Subject to the same imported analytic interface, the same certified seven-point inequality, and the same finite-dimensional Gram-profile estimate, the refinement replaces the coarse global pressure factor $m-1$ by the exact factor $m-6$. With the certified parameters of the predecessor, the resulting numerical lower bound is

$$\liminf_{T \rightarrow \infty} \frac{N_0^s(T, 2T)}{N(T, 2T)} \geq 0.673262375584978050386 \dots,$$

so in particular it exceeds 0.6732623755. This is a same-day research draft, not a peer-reviewed theorem and not a proof of the Riemann hypothesis.

1 Statement and trust boundary

Let $N(T, 2T)$ denote the number of nontrivial zeros of ζ with ordinates in $(T, 2T]$, counted with multiplicity, and let $N_0^s(T, 2T)$ denote the number of simple zeros on the critical line in the same range. Write

$$S = N_0^s(T, 2T), \quad N = N(T, 2T).$$

We import the analytic interface

$$S \geq H_{\text{cert}} N + \text{tr } \Psi(M) - o(N), \tag{1}$$

with

$$H_{\text{cert}} = \frac{672457041414544284}{10^{18}},$$

and the certified seven-point inequality of the predecessor with

$$\varepsilon = \frac{51063}{10^7}, \quad b_r \geq 0, \quad B := \sum_{r=1}^6 b_r = \frac{3}{1150}.$$

We also import the finite-dimensional block estimate

$$\text{tr } \Psi(G) \geq h_m(E), \tag{2}$$

where

$$h_m(E) = \begin{cases} E, & 0 \leq E \leq \frac{m}{m-1}, \\ \frac{E}{m} + 2\sqrt{\frac{m-1}{m}}E - 1, & E \geq \frac{m}{m-1}. \end{cases}$$

These inputs are not reproved here.

Theorem 1. *Subject to the imported inputs above and the same normalization used in the predecessor's shifted-block argument,*

$$\liminf_{T \rightarrow \infty} \frac{N_0^s(T, 2T)}{N(T, 2T)} \geq 0.673262375584978050386 \dots > \frac{6732623755}{10^{10}}.$$

2 Exact local pressure multiplicity

Consider a block of m consecutive simple zeros, with normalized gaps

$$g_1, \dots, g_{m-1}.$$

For each seven-point window starting at $t \in \{0, \dots, m-7\}$, the local pressure contribution is

$$\sum_{r=1}^6 b_r g_{t+r}.$$

After summing all $m-6$ local inequalities, define

$$c_j := \sum_{\substack{0 \leq t \leq m-7 \\ 1 \leq j-t \leq 6}} b_{j-t}, \quad P_B := \sum_{j=1}^{m-1} c_j g_j. \quad (3)$$

The pair terms are handled exactly as in the predecessor, so with E_B denoting the corresponding block off-diagonal energy,

$$E_B + P_B \geq A_m, \quad A_m := \varepsilon(m-6). \quad (4)$$

Lemma 2 (Exact pressure multiplicity). *The coefficients in (3) satisfy*

$$\sum_{j=1}^{m-1} c_j = (m-6)B.$$

Proof. This is a direct double count of pairs consisting of a seven-point window and one of its six internal gap positions:

$$\begin{aligned} \sum_{j=1}^{m-1} c_j &= \sum_{j=1}^{m-1} \sum_{\substack{0 \leq t \leq m-7 \\ 1 \leq j-t \leq 6}} b_{j-t} \\ &= \sum_{t=0}^{m-7} \sum_{r=1}^6 b_r \\ &= (m-6)B. \end{aligned}$$

□

3 Retaining the exact pressure through the defect estimate

Set

$$R_m := h_m(A_m), \quad \eta_m := \frac{R_m}{A_m}.$$

Because h_m is increasing and concave with $h_m(0) = 0$, one has

$$h_m(E) \geq \eta_m E \quad (0 \leq E \leq A_m). \quad (5)$$

If $E_B \geq A_m$, monotonicity and (2) give

$$\mathrm{tr} \Psi(G_B) \geq R_m - o(1).$$

If $E_B < A_m$, then (4) gives $P_B \geq A_m - E_B$, and therefore by (5),

$$\begin{aligned} \mathrm{tr} \Psi(G_B) + \eta_m P_B &\geq \eta_m E_B + \eta_m (A_m - E_B) - o(1) \\ &= R_m - o(1). \end{aligned}$$

Thus in all cases

$$\mathrm{tr} \Psi(G_B) + \eta_m P_B \geq R_m - o(1). \quad (6)$$

The key point is that P_B is retained in its exact position-dependent form; it is not replaced at this stage by the coarser $B \operatorname{span}(B)$.

4 Shifted-block averaging

Partition the ordered simple zeros into disjoint blocks of m consecutive points, and consider all m residue-class shifts of this partition. Up to $O(m)$ endpoint effects, a fixed global gap has the following behavior across the m shifts:

- (i) it is a block boundary exactly once;
- (ii) it occupies each internal position $j = 1, \dots, m-1$ exactly once.

Consequently, by Lemma 2, its total pressure coefficient over all shifted partitions is exactly

$$\sum_{j=1}^{m-1} c_j = (m-6)B. \quad (7)$$

Averaging (6) over the shifted partitions, using spectral pinching for the block defects and the same normalized total-gap relation as in the predecessor, gives

$$\mathrm{tr} \Psi(M) \geq \frac{R_m}{m} S - \eta_m B \frac{m-6}{m} N - o(N). \quad (8)$$

For comparison, replacing each c_j by the uniform upper bound B before the shifted average yields the larger factor $B(m-1)/m$.

5 Assembly and numerical specialization

Substituting (8) into (1) gives

$$S \geq H_{\text{cert}}N + \frac{R_m}{m}S - \eta_m B \frac{m-6}{m}N - o(N),$$

and hence

$$\frac{S}{N} \geq \frac{mH_{\text{cert}} - \eta_m B(m-6)}{m - R_m} - o(1). \quad (9)$$

Since $A_m = \varepsilon(m-6)$,

$$\eta_m B(m-6) = \frac{R_m B}{\varepsilon},$$

so (9) may also be written

$$\frac{S}{N} \geq \frac{mH_{\text{cert}} - R_m B/\varepsilon}{m - R_m} - o(1).$$

A high-precision scan of integer block lengths selects

$$m = 215.$$

Then

$$A_{215} = \frac{51063}{10^7} \cdot 209 = 1.0672167,$$

and

$$R_{215} = 1.06627687661618780546392584286702014367 \dots$$

Substitution into (9) yields

$$\frac{S}{N} \geq 0.67326237558497805038619164619309578448 \dots - o(1).$$

This proves Theorem 1, subject to the imported inputs.

6 Reproducibility

The repository contains two independent executable checks:

- `src/check_multiplicity.py`, which verifies the exact rational identity in Lemma 2 from the published pressure coefficients;
- `src/check_final_bound.py`, which scans the block length and then recomputes the final bound with 100-digit interval arithmetic.

The GitHub Actions workflow runs both checks on every push and pull request.

7 Research status and attribution

This note contributes only the exact pressure-multiplicity retention through shifted-block averaging and the resulting arithmetic refinement. The analytic interface, the local seven-point interval certificate, the re-optimized window, and the sharp finite- m block profile are inherited from the Anthropic / `trmdy` / `sxuff` lineage. If any imported input fails, the numerical bound in this note fails with it.

This manuscript should receive independent mathematical review before being cited as an established theorem.

References

- [1] `sxuff/zeta-positioned-pressure`, <https://github.com/sxuff/zeta-positioned-pressure>.
- [2] `trmdy/zeta-simple-zeros-673137`, <https://github.com/trmdy/zeta-simple-zeros-673137>.
- [3] Anthropic, research artifacts on simple zeros of the Riemann zeta function, 2026.